

Species identity retrieval

TRACK THE SAME ELEMENT BEFORE COMPARING



OPEN-ENDED TASK | attempt before help

Without using any Redox label, map each element from reactant to product.

SELF-GUIDED HELP

H1 ORIENT: identify the evidence target. H2 REPRESENT: use the displayed frame. H3 PRINCIPLE: Match Zn only with Zn²⁺ and Cu in Cu²⁺ only with Cu. H4 FIRST MOVE: write the first evidence statement before any role label.

CANONICAL RESPONSE + CHECK

Zn -> Zn²⁺; Cu²⁺ -> Cu. Identity comes before labels. Check that the same element is tracked and the electron ledger closes.

Construct the oxidation-state lanes

OXIDATION-STATE LANES

Zn: 0 -> +2 (increase)

Cu: +2 -> 0 (decrease)

OPEN-ENDED TASK | attempt before help

Write the before/after oxidation state for each element and describe only the numerical direction.

SELF-GUIDED HELP

H1 ORIENT: identify the evidence target. H2 REPRESENT: use the displayed frame. H3 PRINCIPLE: Elemental Zn/Cu are 0; the ions shown carry +2 for the corresponding element. H4 FIRST MOVE: write the first evidence statement before any role label.

CANONICAL RESPONSE + CHECK

Zn 0 -> +2 increases. Cu +2 -> 0 decreases. Check that the same element is tracked and the electron ledger closes.

Build oxidation / reduction from direction

OXIDATION-STATE LANES

Zn: 0 -> +2 (increase)

Cu: +2 -> 0 (decrease)

OPEN-ENDED TASK | attempt before help

Write the general rule first, then use it to label Zn and Cu²⁺.

SELF-GUIDED HELP

H1 ORIENT: identify the evidence target. H2 REPRESENT: use the displayed frame. H3 PRINCIPLE: Do not use a remembered reagent role. Use increase/decrease. H4 FIRST MOVE: write the first evidence statement before any role label.

CANONICAL RESPONSE + CHECK

Increase -> oxidation, so Zn is oxidised. Decrease -> reduction, so Cu²⁺ is reduced. Check that the same element is tracked and the electron ledger closes.

Construct the electron consequences

ELECTRON LEDGER



2 electrons lost = 2 electrons gained

OPEN-ENDED TASK | attempt before help

Translate each self-change into an electron statement and close the ledger.

SELF-GUIDED HELP

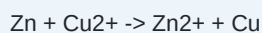
H1 ORIENT: identify the evidence target. H2 REPRESENT: use the displayed frame. H3 PRINCIPLE: The species oxidised loses electrons; the species reduced gains them. H4 FIRST MOVE: write the first evidence statement before any role label.

CANONICAL RESPONSE + CHECK

$\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^{-}$. $\text{Cu}^{2+} + 2\text{e}^{-} \rightarrow \text{Cu}$. Two lost = two gained. Check that the same element is tracked and the electron ledger closes.

Construct the Zn agent role

GOVERNED REACTION



OPEN-ENDED TASK | attempt before help

Build a three-link proof: Zn self-change -> effect on Cu^{2+} -> role.

SELF-GUIDED HELP

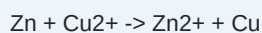
H1 ORIENT: identify the evidence target. H2 REPRESENT: use the displayed frame. H3 PRINCIPLE: Role comes last. Start 'Zn is oxidised and loses electrons...'. H4 FIRST MOVE: write the first evidence statement before any role label.

CANONICAL RESPONSE + CHECK

Zn is oxidised and supplies electrons that reduce Cu^{2+} ; therefore Zn is the reducing agent. Check that the same element is tracked and the electron ledger closes.

Construct the Cu²⁺ agent role

GOVERNED REACTION



OPEN-ENDED TASK | attempt before help

Build the complementary three-link proof for Cu²⁺.

SELF-GUIDED HELP

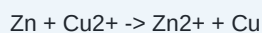
H1 ORIENT: identify the evidence target. H2 REPRESENT: use the displayed frame. H3 PRINCIPLE: Start with Cu²⁺ being reduced / gaining electrons; then connect that to Zn oxidation. H4 FIRST MOVE: write the first evidence statement before any role label.

CANONICAL RESPONSE + CHECK

Cu²⁺ is reduced and accepts electrons from Zn; therefore Cu²⁺ is the oxidising agent. Check that the same element is tracked and the electron ledger closes.

Diagnose the classic reversal

GOVERNED REACTION



OPEN-ENDED TASK | attempt before help

A learner says 'Zn is the reducing agent because Zn gets reduced.' Identify the exact broken link.

SELF-GUIDED HELP

H1 ORIENT: identify the evidence target. H2 REPRESENT: use the displayed frame. H3 PRINCIPLE: Check the Zn lane before discussing the role word. H4 FIRST MOVE: write the first evidence statement before any role label.

CANONICAL RESPONSE + CHECK

Zn does not get reduced: 0 -> +2 is oxidation. Its oxidation supplies electrons for Cu²⁺ reduction; that is why it is the reducing agent. Check that the same element is tracked and the electron ledger closes.

Translate between proof representations

ELECTRON LEDGER



2 electrons lost = 2 electrons gained

OPEN-ENDED TASK | attempt before help

Show how one statement can be written as a state lane, an electron statement, and an agent-role sentence.

SELF-GUIDED HELP

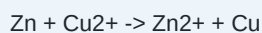
H1 ORIENT: identify the evidence target. H2 REPRESENT: use the displayed frame. H3 PRINCIPLE: Use Zn as the anchor and preserve the same self-change across all three forms. H4 FIRST MOVE: write the first evidence statement before any role label.

CANONICAL RESPONSE + CHECK

$\text{Zn} 0 \rightarrow +2$ | $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^{-}$ | Zn is oxidised and acts as the reducing agent. Check that the same element is tracked and the electron ledger closes.

Teach back without the page

GOVERNED REACTION



OPEN-ENDED TASK | attempt before help

From $\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$ alone, reconstruct the full Redox proof in your own layout.

SELF-GUIDED HELP

H1 ORIENT: identify the evidence target. H2 REPRESENT: use the displayed frame. H3 PRINCIPLE: Minimum chain: identity -> state direction -> electron consequence -> role -> independent check. H4 FIRST MOVE: write the first evidence statement before any role label.

CANONICAL RESPONSE + CHECK

$\text{Zn } 0 \rightarrow +2$ oxidised, loses $2e^-$, reducing agent. $\text{Cu } +2 \rightarrow 0$ reduced, gains $2e^-$, oxidising agent. Electron loss equals gain. Check that the same element is tracked and the electron ledger closes.